Test 7 – Grade 9

Name: _____
Grade / Class: 9

Section A: English Language & Literature (Questions 1–15)

1. **Read the passage: *"Sarah decided to research local wildlife. She observed birds, recorded their behavior, and sketched their nests."***
   Question: What was Sarah's main activity during her research?
2. **List two methods Sarah used to study the birds.**
3. **Explain why recording behavior is important in wildlife research.**
4. **Use the word *sketch* in a meaningful sentence.**
5. **Rewrite the sentence in passive voice: *"Sarah observed the birds carefully."***
6. **Correct the grammatical error: *"Neither the students nor the teacher were aware of the rules."***
7. **Identify the part of speech of the underlined word: *"The birds flew gracefully."***
8. **Write a formal letter to the principal requesting permission for a field trip to a nature reserve (80–100 words).**
9. **Compose a short story on the theme: *"A Mysterious Discovery"* (50–70 words).**
10. **Define the literary term *simile* and give an original example.**
11. **Differentiate between first-person and third-person narration with examples.**
12. **Summarize the passage in 40 words.**
13. **Change the sentence into a question: *"The students completed their assignments on time."***
14. **Write a paragraph describing a traditional festival in your country (60–80 words).**
15. **Explain the theme of the following sentence: *"Hard work and patience lead to success."***

Section B: Mathematics (Questions 16–35)

16. **Solve for x: $5x - 7 = 18$**
17. **Factorize: $x^2 + 9x + 20$**
18. **Simplify: $(2x^2y^3)(3xy^2)$**
19. **Solve the system of equations:**
    $2x + 3y = 12$
    $x - y = 1$
20. **Calculate the area of a triangle with base 10 cm and height 6 cm.**

21. Solve: $4(3x - 5) = 2x + 14$
22. Expand: $(x + 4)(x - 3)$
23. Find the HCF of 48 and 180.
24. A car travels 360 km in 4 hours. Find its average speed.
25. Solve for y: $3y/4 - 5 = 7$
26. Find the roots of $x^2 - 5x + 6 = 0$.
27. Calculate the perimeter of a rectangle with length 12 cm and width 7 cm.
28. Simplify: $2a^2b^3 \times 3ab^2$
29. A bag contains 5 red, 7 blue, and 8 green balls. Probability of picking a red ball?
30. Solve: $2(x - 3) + 5 = 3x - 1$
31. Convert 3.75 into a fraction.
32. The sum of angles in a polygon is 1080°. Find the number of sides.
33. Solve for x: $7(x - 2) = 3x + 10$
34. The ratio of boys to girls in a class is 3:4. If there are 21 boys, how many girls?
35. Simplify: $(5x^2y^3)^2 \div (x^3y)$

Section C: Physics (Questions 36–50)

36. Define Newton's Second Law of Motion.
37. A force of 20 N acts on a mass of 4 kg. Find the acceleration.
38. Explain the difference between speed and velocity.
39. A car accelerates from 0 to 60 m/s in 10 seconds. Find acceleration.
40. Define potential energy with an example.
41. A 2 kg object is lifted to a height of 5 m. Calculate its potential energy ($g = 10\ m/s^2$).
42. Explain the principle of conservation of momentum.
43. Define work in physics and write its formula.
44. A force of 50 N moves an object 2 m. Calculate work done.
45. Differentiate between scalar and vector quantities with examples.
46. State the law of reflection.
47. Explain the difference between conduction, convection, and radiation.
48. A 5 Ω resistor carries 2 A of current. Find the voltage across it.
49. Define power in electrical circuits.
50. Calculate the power consumed by a device using 120 V and 3 A.

Section D: Chemistry (Questions 51–65)

51. Define an acid and give an example.
52. Write the chemical formula of calcium carbonate.
53. Balance the equation: $H_2 + O_2 \rightarrow H_2O$

54. Define a base and provide an example.
55. Explain the difference between elements, compounds, and mixtures.
56. Write the electron configuration of oxygen.
57. Define oxidation and reduction.
58. Name the products of the reaction between hydrochloric acid and sodium hydroxide.
59. Classify the following as physical or chemical change: *Melting of ice.*
60. Explain the periodic table and its significance.
61. Give the formula of sulphuric acid.
62. Define the term *isotope* with an example.
63. Explain the difference between metals and non-metals.
64. Write the chemical equation for rusting of iron.
65. Describe the pH scale and its importance.

Section E: Biology (Questions 66–80)

66. Define photosynthesis.
67. Write the chemical equation for photosynthesis.
68. Explain the function of mitochondria.
69. Differentiate between plant and animal cells.
70. Describe the structure of the human heart.
71. What is osmosis? Give an example.
72. Explain the role of enzymes in digestion.
73. Name two diseases caused by bacteria.
74. Define habitat and give an example.
75. Explain the difference between vertebrates and invertebrates.
76. Describe the process of respiration in humans.
77. Name the three types of muscles in the human body.
78. Define immunity and its types.
79. Explain the nitrogen cycle.
80. What is the function of chlorophyll in plants?

Section F: Social Studies / History / Geography (Questions 81–90)

81. Define democracy.
82. Name three major rivers of Asia.
83. Explain the causes of World War I.
84. Define climate change.
85. Identify the capital of Brazil.

86. Explain the concept of human rights.
87. Name two major mountain ranges in the world.
88. Describe the greenhouse effect.
89. Identify three ancient civilizations.
90. Explain the difference between latitude and longitude.

Section G: Computer / ICT (Questions 91–95)

91. Define a programming language.
92. Name two types of computer memory.
93. What is the function of an operating system?
94. Explain the difference between hardware and software.
95. Define the term *database*.

Section H: Islamiyat / Religious Studies (Questions 96–100)

96. Name the five pillars of Islam.
97. Explain the importance of Salah in daily life.
98. Define Zakat and its significance.
99. What is the Quran, and why is it important in Islam?
100. Explain the story of Prophet Yusuf (Joseph) and the main moral lesson from it.